Thursday 01/22 Notes

Surjective means: $f: D \to C$ if for every $y \in C$ there exists x in D such that $f(x) = y$

For every $= \forall$
$\exists =$ exists

"Everything gets mapped to"

is surjective but not injective

Claim: $f(x) = x^2$ as a function is surjective from $\{0,1\}$ to $\{0,1\}$

Proof:

Let $y \in \{0,1\}$

If $y = 0$ then we see $f(0) = 0^2 = 0$

If $y = 1$, then we see $f(1) = 1^2 = 1$

Thus we see $f: \{0,1\}$ to $\{0,1\}$ is surjective

Proof: We compute

$$0 \xrightarrow{f} 0$$
$$1 \xrightarrow{f} 1$$

Proof: Note $0^2 = 0$ and $1^2 = 1$, Thus f is surjective

4b from HW 0) $f(x) = x^2 + 1$ $\quad f^3$ means $f(f(f(x)))$

$= ((x^2+1)^2+1)^2+1$

The 3 Primitive Terms are $N, 0, \sigma$ ← Sigma

Axiom 1: N is a set
0 is an element of this set $\quad 0 \in N$
σ is a function from $N \to N$ $\quad \sigma: N \to N$

Axiom 2: The image $\sigma: N \to N$ does not contain 0

Def: If $f: D \to C$ the image of f, denoted $\mathrm{Im}(f)$ or $f(D)$, is the set $\{y \in C: \exists x \in D \text{ with } f(x) = y\}$ $\quad 0 \notin \sigma(N)$

Axiom 3: The function $\sigma : \mathbb{N} \to \mathbb{N}$ is injective
If $\sigma(x) = \sigma(y)$, then $x = y$

injective visual
graphing

$A \Rightarrow B$ is logically equivalent to not $B \Rightarrow A$
not

If $x \neq y$ then $\sigma(x) \neq \sigma(y)$

Axiom 4: If $S \subseteq \mathbb{N} \wedge 0 \in S$
S is a subset of N including 0 is included in S
i) $0 \in S$
ii) if $n \in S$, then $\sigma(n) \in S$
then $S = \mathbb{N}$

i) Base case

If the nth domino has fallen, then the $n^{th}+1$ will fall

ii) Induction Step

Then the result holds for all n
(thinking $\mathbb{N}$)

Exercise 1: Go through axioms and convince yourself they hold for natural #s $(1, 2, \ldots)$